

Dmitrii Petukhov

Senior/Lead Software Engineer – Java, AWS, AI

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SUMMARY

20+ years building software across telecom, industry, fintech, and retail. Eight years at Amazon architecting cloud-native microservices, low-code automation platforms, and developer tools. Untangling complexity debt in large systems. Mentored 50+ engineers, won two EU-wide innovation awards. Co-organizer of Luxembourg's largest engineering community. Now building and shipping AI products solo.

EXPERIENCE

Senior Software Engineer & Architect

Self-employed Mar 2026 - Present

- Shipped [Career Repo](#), a local-first job-hunt copilot that finds job opportunities in your inbox, scores them, and drafts cover letters in your voice. Presented at Lux TechPulse in June 2026.
- Building [Paragrapher](#), an AI-powered book translator for indie authors. Iterative refinement converges each line toward human-grade quality, with per-line scoring and glossary management. In production use.
- Building [Plankton](#), an AI-powered embeddable landscape designer with a back-office for professional contractors. Generates CAD-ready plans and material lists. One design partner signed, recruiting more.

Technologies & skills: Java 21, TypeScript, Python, Spring Boot 3, Spring AI, FastAPI, React 19, Tailwind 4, PostgreSQL 18, SQLite, sqllite-vec, AWS ECS, RDS, CDK, Docker, Anthropic Claude API, Claude Agent SDK, OpenAI API, Google Gemini API, xAI Grok API, sentence-transformers embeddings, agentic workflows, LLM prompt engineering, Yjs CRDT, collaborative and single-user architectures, computational geometry, Stripe

Senior Software Engineer, Lead

Amazon Feb 2018 - Feb 2026

- Supported 8 marketplace launches across EMEA and APAC, impacting 1,000+ services and enabling 100M+ new customers
- Designed and delivered a dynamic configuration platform for 600+ services, reducing country launch coding effort from 50 engineer-weeks to one day
- Introduced a meta-modeling technique to document systems at scale, adopted by 10+ teams
- Reduced operational incidents by 80% through enhanced monitoring, emergency automation, and production readiness reviews
- Mentored 50+ engineers through workshops, design reviews, and 1:1 coaching
- Honored with two Amazon EU awards for innovation and automation

Technologies & skills: Java, Kotlin, Python, Spring Boot, REST, SQL, PostgreSQL, AWS ECS, Lambda, S3, RDS, DynamoDB, SQS, SNS, CloudWatch, CDK, Apache Spark, ETL, data lake architecture, distributed batch processing, cumulative data ingestion, financial data pipelines at scale, EMR, Redshift, Pub/Sub, Docker, CloudFormation, React/TypeScript, meta-modeling, architecture decisions, Kanban

Senior Software Engineer

Upwork Feb 2017 - Feb 2018

- Developed low-latency APIs in a 300+ microservice system serving 14M+ users on Upwork's web and mobile platforms
- Increased deployment speed 15x by optimizing build scripts and tuning CI/CD pipelines
- Accelerated new service creation to one day via code-generated GraphQL schemas and declarative REST APIs
- Standardized end-to-end quality assurance for 10+ mobile API services through multi-step scenario tests

Technologies & skills: Java, Groovy, Spring Boot, Dropwizard, Netflix Hystrix, Archaius, Apache Shiro, GraphQL, REST, gRPC, Kafka, event-driven architecture, PostgreSQL, Redis, Elasticsearch, JOOQ, Flyway, AWS, Gradle, Jenkins, Docker, ELK, Prometheus, Grafana, microservices, CI/CD optimization, blue/green deployments, gradual rollouts, automated rollbacks

Senior Software Engineer

ACI Worldwide Feb 2015 - Feb 2017

- Developed payment solutions powering over 5,000 banks and billers
- Refactored a large monolithic backend into modular services and modernized legacy UIs
- Created a QA automation tool reducing test data preparation time by 10x
- Served as Certified Scrum Master for two agile teams

Technologies & skills: Java, Spring Boot, Spring Data JPA, REST, ExtJS, Oracle, SQL, Liquibase, JUnit, Mockito, Jenkins, deployment automation, release engineering, payment systems, backend modularization, QA automation, agile delivery, Certified Scrum Master

EARLY CAREER

Desktop applications for telecom and industrial domains 2007 - 2015

Math-heavy and graphics-heavy, single-user. Delphi, FireMonkey, WebGL, GDI+. Integration with lab and POS equipment: COM ports, cash validators, barcode scanners, fiscal printers, touchscreen self-service kiosks.

EDUCATION

Master's in Applied Mathematics and Computer Science

Kazan National Research Technical University Sep 2002 - Jun 2007

PERSONAL TENETS

Business outcomes first · There are no juniors, only those not trusted yet · Know your tools, and their trade-offs